SR.IIT_*CO-SC(MODEL-A& APEX) GTM-14 Date: 08.01.2025
Time: 3.00Hrs. JEE MAINS -2025 Max. Marks: 300

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
C	D	A	C	C	C	C	A	A	D
11	12	13	14	15	16	17	18	19	20
A	C	C	A	D	B	C	B	A	A
21	22	23	24	25					
3	49	2	12	2					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	B	D	A	B	C	C	C	A	D
36	37	38	39	40	41	42	43	44	45
B	C	C	A	D	B	C	B	C	B
46	47	48	49	50					
20	5	50	416	446					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	B	A	B	C	D	C	B	C	C
61	62	63	64	65	66	67	68	69	70
B	A	C	D	A	D	C	B	A	A
71	72	73	74	75					
2	4	10	2	25					

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HINTS
PHYSICS

01. Energy present in volume V $= \dfrac{IV}{c}$

So, number of photons in that volume $N = \dfrac{IV\lambda}{c^2 h} = \dfrac{(100)(10^{-9})(2640\times10^{-10})}{9\times10^{16}\times6.626\times10^{-34}}$

$\Rightarrow \quad N = 442.7$

02. Both the Diodes act like short. Thus, $i_1 = 0$ and $i_2 = \dfrac{10}{2000} = 5mA$

03. $\dfrac{\Delta R}{R}\times100 = \dfrac{\Delta A}{A}\times100 + 3\dfrac{\Delta C}{C}\times100 + 2\dfrac{\Delta B}{B}\times100$

$\dfrac{\Delta R}{R}\% = 0.25\% + 0.2\% + 0.75\% = 1.2\%$

04. $P_{avg} = \dfrac{E_{rms}^2}{Z}\cos\phi$

05. $PV = nRT \quad \Rightarrow \quad n = \dfrac{PV}{RT}$

$n_1 = \dfrac{(2V)(2P)}{R(2T)}$ and $n_2 = \dfrac{(V)(P)}{R(T)}$

So, $n_1 = 2n_2$

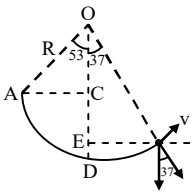
06. $v = \omega\sqrt{A^2 - x^2}$

$\therefore (50\pi) = (10\pi)\sqrt{(10)^2 x^2}$

Or $x = \pm 8.66cm$

Therefore distance between those two points,

$d = 2|x| = 17.32cm$

07. 

In figure

OC = R cos 53° = $\dfrac{3R}{5}$; OE = R cos 37° = $\dfrac{4R}{5}$

So CE = $\dfrac{R}{5}$

From A to B,

$\dfrac{1}{2}mv^2 = mg\dfrac{R}{5}$ [Energy conservation]

v = $\sqrt{\dfrac{2gR}{5}}$

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Let radius of curvature at B is r

$g\cos 37^\circ = \frac{v^2}{r}$

$r = \frac{v^2}{g\cos 37^\circ} = \frac{2gR}{5\times 10\times 4/5} = \frac{R}{2}$

08. $1/v+1/u=1/f$

09. $r \propto \frac{\sqrt{m}}{q}$

10.

$H = -kA\frac{dT}{dx}$

$\Rightarrow \int_0^L Hdx = -aA\int_{4T_0}^{T_0} TdT \Rightarrow HL = aA\frac{15T_0^{\,2}}{2}$

$\Rightarrow \int_0^{\frac{L}{2}} Hdx = -aA\int_{4T_0}^{T} TdT \Rightarrow \frac{HL}{2} = aA\frac{16T_0^{\,2}-T^2}{2}$

$\Rightarrow \frac{15T_0^{\,2}}{2} = 16T_0^{\,2} - T^2 \Rightarrow T = \sqrt{\frac{17}{2}}T_0$

11. $\frac{1}{2}\times\frac{4}{3}\pi R^3\times v^2 = 4\pi T\left(\frac{R^3}{r} - R^2\right)$

$\frac{1}{6}R^3\rho\upsilon^2 = TR^3\left(\frac{1}{r}-\frac{1}{R}\right)$

$\frac{\rho\times\upsilon^2}{6} = T\left(\frac{1}{r}-\frac{1}{R}\right)$

$\Rightarrow \upsilon^2 = \frac{6T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)$

$\Rightarrow \upsilon = \sqrt{\frac{6T}{\rho}\left(\frac{1}{r}-\frac{1}{R}\right)}$

12. If A moves down by a_1 w.r.t B and B moves to the right by a_2 then the constraint relation gives $a_1 = a_2 = a$.

For A : $N = ma; mg - T = ma$

For B : $T - N = ma \Rightarrow T = 2ma$

$\Rightarrow a = \frac{g}{3}$

$\Rightarrow \vec{a}_{AO} = \vec{a}_{BO} + \vec{a}_{AB} = \frac{g}{3}\hat{i} - \frac{g}{3}\hat{j}$

13. Elongation $\Delta = \frac{WL}{E}$

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On either side of the wire, tension is W and length is $\frac{L}{2}$

$$\Delta\ell = \frac{\left(\frac{WL}{2}\right)}{AE}$$

$$= \frac{WL}{2AE}$$

$$\frac{L}{2}$$

14. Geo-satellite means whose time period is

T = 24 hr & from law

$$\frac{T_2^2}{(24)^2} = \frac{(R+2.5R)^3}{(R+6R)^3}$$

$$T_2 = \sqrt{\frac{24\times 24}{8}} = 6\sqrt{2} \text{ hr.}$$

15. $0 = 2mm, u = -20cm$

$$f = \frac{R}{2} = \frac{40}{2} = 20cm$$

From mirror formula,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{20} = \frac{1}{v} + \frac{1}{-20}$$

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{20}$$

$$\Rightarrow v = 10cm$$

$$\therefore \frac{I}{0} = \frac{v}{u}$$

$$\frac{I}{2} = \frac{10}{20}$$

$$\Rightarrow I = 1mm$$

Height of image = 1 mm

16. $\frac{10R}{500} = 2$

17. $v(t) = \frac{dx}{dt} = \frac{d}{dt}\left(u(t-2) + \alpha(t-2)^2\right)$

Using the product and chain rules:

$u(t) = u + 2a(t-2)$

To find the initial velocity at t=0:

v(0)=u+2a(0−2)=u−4a

a(t)=dv/dt=d/dt((u+2a(t−2)))=2a

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18. $Q = nC\Delta T = C\left(2T_0 - T_0\right)$

$\frac{P}{T^2} = \text{constant} \Rightarrow PV^2 = \text{constant} \Rightarrow C = C_V + \frac{R}{1-2} = \frac{3R}{2} - R = \frac{R}{2}$

$\Rightarrow Q = \frac{RT_0}{2}$.

19. $r_{n+1} - r_n = r_{n-1} \Rightarrow \frac{r_0(n+1)^2}{Z^2} - \frac{r_0 n^2}{Z^2} = \frac{r_0(n-1)^2}{Z^2} \Rightarrow n = 4$

$E_4 = -13.6\frac{Z^2}{4^2} = -0.85\ Z^2$.

20. When light ray goes from denser to rarer medium ($i.e.$, more μ to less μ) it deviates away from the normal while if light ray goes from rarer to denser medium ($i.e.$ less μ more μ) it bend towards the normal.
This property is satisfied by the ray diagram (i) only

21. Applying COAM about the centre of the plate:

$$mua = \left(m\left(\frac{\sqrt{5}a}{2}\right)^2 + m\left(\frac{(2a)^2 + (a)^2}{12}\right)\right)\omega \Rightarrow \omega = \frac{3u}{5a}$$

22. B=0.2 T,
q=1.0×10^{-5} C,
m=0.01 g=1×10^{-5} kg
Force due to magnetic field = Gravitational force of attraction
So, qVB=mg
$\Rightarrow 1\times10^{-5}$ V$\times 2\times10^{-1} = 1\times10^{-5}\times 9.8$
V=4.9×10=49m/s

23. $T_1 = \frac{\pi}{6\omega}$

$T_2 = \frac{\pi}{3\omega}$

24. A fully charged capacitor draws no current. If the capacitor is removed from the circuit, we can distribute the current and find the potential difference across each resistance.

25. $n_1\left(\frac{V}{4l}\right) - n_2\left(\frac{V}{2l}\right) = 4$

It l is doubled, beat frequency will remain $\frac{4}{2} = 2Hz$

CHEMISTRY

26. Conceptual

27. $K = \frac{2.303}{20}\log\frac{100}{90}$: $K = \frac{2.303}{t_{19\%}}\log\frac{100}{81}$

$\frac{2.303}{20}\log\frac{10}{9} = \frac{2.303}{t_{19\%}}\log\left(\frac{10}{9}\right)^2$

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$$\frac{1}{20}\log\frac{10}{9}=\frac{2}{t_{19\%}}\log\frac{10}{9}$$

$$t_{19\%}=40\,\text{min}$$

28. No of spectral lines given by H = $\frac{5\times4}{2}=10$

No of spectral lines given by He+ = $\frac{4\times3}{2}=6$

No of spectral lines given by Li2+ = $\frac{3\times2}{2}=3$

However $\lambda_{2\to1}(H)=\lambda_{4\to2}(He^{2+})$

Total no of spectral lines = 10 + 6 + 3 – 1 = 18

29. $\left[Ca^{+2}\right]_{final}=\frac{10^{-2}\times V}{2V}=5\times10^{-3}$

$\left[F^{-}\right]_{final}=\frac{10^{-3}\times V}{2V}=5\times10^{-4}$

$K_{I_P}=\left[Ca^{+2}\right]\left[F^{-}\right]^2$

$=\left[5\times10^{-3}\right]\left[5\times10^{-4}\right]^2$

$=125\times10^{-11}$

$K_{I_P}=1.25\times10^{-9}$

$K_{I_P}>K_{S_P}$ (ppt will form)

30. Only a, c, d correct
31. Mn_2O_7 : multiple bonds

$AgNO_3\xrightarrow{\Delta}Ag\downarrow+NO_2\uparrow+O_2\uparrow$

$\overset{(+5)}{V}\ O_2^+<\overset{(+6)}{Cr}_2O_7^{2-}<\overset{(+7)}{Mn}O_4^-$ When I^- is oxidized by MnO_4^- in alkaline medium, I^- converts into IO_3^-

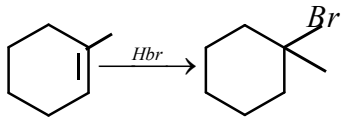
32. Atomic radius of noble gas is highest in the respective period
33. Conceptual
34. All are not isotonic to each other.
35. Benzoic acid is stronger than phenols
36. No. of chiral centre =2

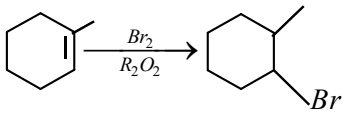
No. of optical isomers = 2^2=4

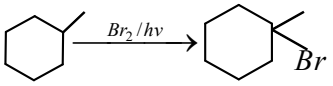
37. Clemmenson reduction acid sensitive group $-OH$ may undergo effect

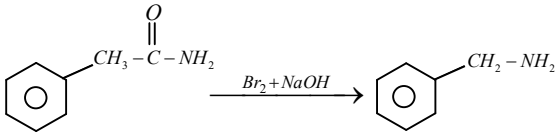
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38.

1) 

2) 

3) 

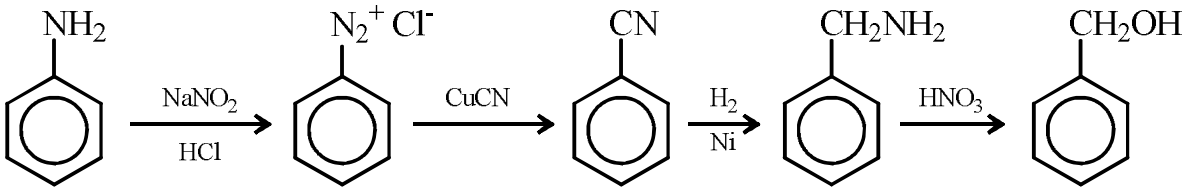
4) 

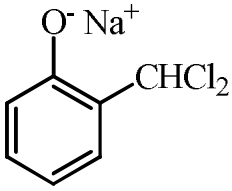
39. $\mu_{cal} = q \times d$

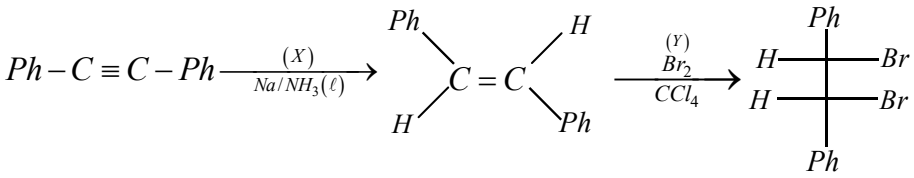
$= \left(1.60 \times 10^{-19} C\right)\left(1.41 \times 10^{-10} m\right)$

$= 2.26 \times 10^{-29} Cm$

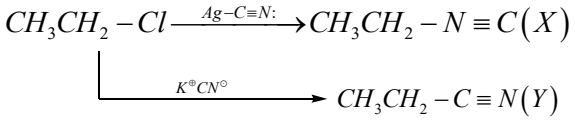
The percentage ionic character $= \dfrac{2.60 \times 10^{-30} Cm}{2.26 \times 10^{-29} Cm} \times 100 = 11.5\%$

40. 

41. The intermediate is 

42. 

43. Kjeldahl's method is not applicable for nitro, nitro so compounds and for compounds is which N is present in the ring

44. 

45. H_2S is a week acid, its dissociation controlled by HCl and NH_4OH.

46. Conceptual

47. Pressure, $p_1 = 20atm$

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Pressure, $p_2 = 50atm$

Temperature, $T_1 = 27°C + 273°C = 300K$

Temperature, $T_2 = 327°C + 273°C = 600K$

Reaction,

$2NH_{3(g)} \rightleftharpoons N_{2(g)} + 3H_{2(g)}$

After dissociation $1-\alpha$ $\quad \frac{\alpha}{2} \quad \frac{3\alpha}{2}$

Total moles $= 1+\alpha$

Ideal gas equation,

$\Rightarrow PV = nRT$

$\Rightarrow \frac{20V}{50V} = \frac{1}{n}\frac{R}{R}\frac{300}{600}$

$\Rightarrow n = 1.25$

$\Rightarrow \alpha = 0.25$

$\Rightarrow \% = 25$

i.e., $x = 5$

48. $gr.Eq$ of $Fe^{+2} = grEq$ of $KMnO_4$

$$\frac{wt}{393} \times 1 = 0.1 \times \frac{50}{1000}$$

$$wt = \frac{1.0 \times 50 \times 392}{1000}$$

$$= 1.96$$

$$\frac{1.96}{3.92} \times 100 = 50\%$$

49. $CH_{4(g)} \rightarrow C_{(g)} + 4H_{(g)}; \quad \Delta H = ?$

From the given information on solving we get

$CH_{4(g)} \rightarrow C_{(g)} + 4H_{(g)}; \ \Delta H = 1665 \ kJ/mol$

Then $C-H$ bond energy $= \frac{1665}{4} kJ/mol$

$= 416 \ kJ/mol$

50. When Hg is used as the cathode, Na^+ will be reduced to form sodium amalgam, Na-Hg.

At cathode: $Na^+ + 1e^- + Hg \rightarrow Na-Hg$

2.0 moles of $NaCl$ presence in the solution contains 2 moles of Na^+ in 0.5 liters of 4 M solution. Hence two moles of Na-Hg are formed.

The mass of 2 moles of Na-Hg=2(23+200)=446g.

MATHEMATICS

51. $\lim_{n\to\infty} (-1)^{n+1} \sin\left(\sqrt{c}\sqrt{n^2+0.5n+1}\right); n \in N$

$\sin(n\pi - \theta) = (-1)^{n+1}\sin\theta$

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$$\Rightarrow \lim_{n\to\infty} \sin\left(n\pi - \pi\sqrt{n^2+0.5n+1}\right)$$

$$\Rightarrow \lim_{n\to\infty} \sin\left(\pi\left[\frac{\left(n-\sqrt{n^2+0.5n+1}\right)\left(n+\sqrt{n^2+0.5n+1}\right)}{n+\sqrt{n^2+0.5n+1}}\right]\right)$$

$$\Rightarrow \lim_{n\to\infty} \sin\left(\pi\left[\frac{\cancel{n^2} - \cancel{n^2} - 0.5n - 1}{n+\sqrt{n^2+0.5n+1}}\right]\right) \text{ dividing by n}$$

$$\Rightarrow \lim_{n\to\infty} \sin\left(\pi\left[\frac{-0.5-\frac{1}{n}}{1+\sqrt{1+0.5/n+\frac{1}{n^2}}}\right]\right)$$

$$\Rightarrow \sin\left(\pi\left(-\frac{1}{4}\right)\right) \Rightarrow \frac{-1}{\sqrt{2}}$$

52. Clearly, $f(x)$ in decreasing just before $x=3$ and increasing after $x=3$. For $x=3$ to be the point of local minima, $f(3) < f(3^-)$.

or $-15 < 12 - 27 + \ln\left(a^2-3a+3\right)$

or $a^2-3a+3>1$

or $a \in (-\infty,1)\cup(2,\infty)$

53. $T_{r+1} = {}^{15}C_r.2^{\frac{15-r}{5}}.5^{\frac{r}{3}}$

$r = 0,15$

$2^3.5^0 + 2^0.5^5$

54. $$\int \frac{\left(\sin^{\frac{3}{2}}\theta + \cos^{\frac{3}{2}}\theta\right)d\theta}{\sqrt{\sin^3\theta\cos^3\theta\sin(\theta+\alpha)}}$$

$$\int \frac{\left(\tan^{\frac{3}{2}}\theta + 1\right)d\theta}{\sqrt{\sin^3\theta(\sin\theta\cos\alpha+\cos\theta\sin\alpha)}}$$

$$\int \frac{\left(\tan^{\frac{3}{2}}\theta + 1\right)d\theta}{\cos^2\theta\tan\theta\sqrt{\tan\theta(\cos\alpha\tan\theta+\sin\alpha)}}$$

Let $a=\cos\alpha, b=\sin\alpha$

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Put $\tan\theta = t \quad \Rightarrow \quad \sec^2\theta d\theta = dt$

$$\int \frac{\left(t^{\frac{3}{2}}+1\right)dt}{t^{\frac{3}{2}}(at+b)^{\frac{1}{2}}} = \frac{2}{a}\sqrt{at+b} + \int \frac{dt}{t^{\frac{3}{2}}(at+b)^{\frac{1}{2}}}$$

Put $t = \frac{b}{a}\tan^2 y \quad \Rightarrow \quad dt = 2\frac{b}{a}\tan y \sec^2 y\, dy$

$$= \frac{2}{a}\sqrt{at+b} + \frac{2a^{\frac{1}{2}}}{b}\int \frac{\cos y}{\sin^2 y}dy$$

$$= \frac{2}{a}\sqrt{at+b} - \frac{2a^{\frac{1}{2}}}{b}\frac{1}{\sin y} + c$$

$$= \frac{2}{a}\sqrt{a\tan\theta + b} - \frac{2\sqrt{a}}{b}\frac{\sqrt{a\tan\theta+b}}{\sqrt{a\tan\theta}} + c$$

$$= \frac{2}{a}\sqrt{a\tan\theta+b} - \frac{2}{b}\sqrt{a+b\cot\theta} + c$$

$$= \frac{2}{\cos\alpha}\sqrt{\cos\alpha\tan\theta+\sin\alpha} - \frac{2}{\sin\alpha}\sqrt{\cos\alpha+\sin\alpha\cot\theta} + c$$

55. Equation of line parallel to $3x - y = 7$ is $3x - y = \lambda$

$\because (1,2)$ *is on it.*

$\therefore \lambda = 1$

Now, point of intersection of $3x - y = 1$ and $x + y + 5 = 0$ is $(-1,-4)$.

$\therefore$ Distance between $(1,2)$ *and* $(-1,-4) = \sqrt{2^2+6^2} = \sqrt{40}$

56. Eccentricity of ellipse $= \frac{3}{4}$, *Focii* $= (\pm 3, 0)$

Eccentricity and Foci of hyperbola be $\frac{1}{12}\sqrt{144+\alpha}$ *and* $\left(\pm\frac{12}{5}\frac{\sqrt{144+\alpha}}{12}, 0\right)$

Since foci coincide $\therefore 3 = \frac{1}{5}\sqrt{144+\alpha} \Rightarrow \alpha = 81$, Length of latus rectum $= \frac{2b^2}{a} = \frac{27}{10}$

57. $\overrightarrow{OC} = m\overrightarrow{OA} + n\overrightarrow{OB}$

$\vec{c} = m\vec{a} + n\vec{b}$

Given $|\vec{a}| = 1, |\vec{b}| = 1, |\vec{c}| = \sqrt{2}, \tan\alpha = 7$

Now take dot of equation (1) with $\vec{a}$ and $\vec{b}$ to get

$m = 5/4; \quad n = 7/4$

$\therefore m + n = 3$

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58. $P''(x)$ is constant $\Rightarrow P(x)$ is quadratic

59. $x^{2x}-2x^{x}\cot y-1=0 \qquad when\ x=1$

$1-2\cot\left(y(1)\right)-1=0 \quad \Rightarrow \cot\left(y(1)\right)=0; y(1)=\frac{\pi}{2}$

Now, we have $x^{2x}-2x^{x}\cot y-1=0$

Differentiate both sides with respect to x. $\qquad when\ x=1$

$2(\ln x+1)x^{2x}-2x^{x}(\ln x+1)\cot y+2x^{x}\operatorname{cosec}^{2}y\frac{dy}{dx}=0$

$2-0+\operatorname{cosec}^{2}\frac{\pi}{2}.y'('1)=0$

$y'(1)=-1$

60. $2xy\frac{dy}{dx}-y^{2}dx=(x^{2}+1)dx$

$\frac{x.d(y^{2})-y^{2}.d(x)}{x^{2}}=\frac{(x^{2}+1)dx}{x^{2}}$

$d\left(\frac{y^{2}}{x}\right)=\left(1+\frac{1}{x^{2}}\right)dx$

$\frac{y^{2}}{x}=x-\frac{1}{x}$

$y^{2}=x^{2}-1$

$x^{2}-y^{2}=1$

61. Required no.of ways = coeff of

$x^{30}\ in\left(x^{2}+x^{3}+....+x^{8}\right)^{5}=coeff\ of\ x^{30}\ in\left(\frac{x^{2}(1-x^{7})}{1-x}\right)^{5}$

$=coeff\ of\ x^{20}\ in(1-x^{7})^{5}(1-x)^{-5}={}^{24}C_{20}-5^{17}C_{13}+10^{10}C_{6}=826$

62. $8b^{3}-a^{3}-c^{3}=6abc$

$a^{3}+c^{3}+(-2b)^{3}=3ab(-2b)$

$\Rightarrow a+c-2b=0$ (or) $a=c=-2b$

63. $Let\ A\ is\ (1,0), B\ is\ \left(\frac{1}{2},\frac{\sqrt{3}}{2}\right) and\ \ P\ is(x,y)$

$\Rightarrow \overrightarrow{OP}=(2-t)\overrightarrow{OA}+t\overrightarrow{OB}, t\in R$

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$$\Rightarrow x\hat{i}+y\hat{j}=(2-t)\hat{i}+t\left(\frac{\hat{i}}{2}+\frac{\sqrt{3}\hat{j}}{2}\right)$$

$$\therefore x=2-\frac{t}{2};y=\frac{\sqrt{3}t}{2}$$

$$\left|\overrightarrow{AP}\right|=\left|(x-1)\hat{i}+y\hat{j}\right|=\sqrt{(x-1)^2+y^2}=\sqrt{t^2-t+1}\geq\frac{\sqrt{3}}{2}$$

64. $\forall x\in(-\infty,-1]$

$$f(x)=\sin^{-1}\left(\frac{2x}{1+x^2}\right)-2\tan^{-1}x$$

$$=-\pi-2\tan^{-1}x-2\tan^{-1}x$$

$$=-\pi-4\tan^{-1}x$$

$$g(x)=\sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)+4\tan^{-1}x$$

$$=\frac{\pi}{2}-\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)+4\tan^{-1}x$$

$$=\frac{\pi}{2}-\left(-2\tan^{-1}x\right)+4\tan^{-1}x$$

$$=\frac{\pi}{2}+6\tan^{-1}x$$

$$Now,\quad f(x)-g(x)=\frac{-3\pi}{2}-10\tan^{-1}x$$

$$\forall x\in(-\infty,-1],\ \tan^{-1}x\in\left(\frac{-\pi}{2},\frac{-\pi}{4}\right]$$

$$So,\quad (f(x)-g(x))\in\left[\pi,\frac{7\pi}{2}\right)$$

65. Let $A\equiv(x_1,y_1),B\equiv(x_2,y_2),C\equiv(x_3,y_3),D\equiv(x_4,y_4)$

$$Given\ x_1^2+x_2^2+x_3^2+x_4^2+y_1^2+y_2^2+y_3^2+y_4^2-2x_1x_3-2x_2x_4-2y_2y_3-2y_1y_4\leq 0$$

$$or\ (x_1-x_3)^2+(x_2-x_4)^2+(y_2-y_3)^2+(y_1-y_4)^2\leq 0$$

$$or\ \frac{x_1+x_2}{2}=\frac{x_3+x_4}{2}\ and\ \frac{y_1+y_2}{2}=\frac{y_4+y_3}{2}$$

Hence , AB and CD bisect each other .

Therefore , ABCD is a parallelogram , Also

$$AB^2=(x_1-x_2)^2+(y_1-y_2)^2$$

$$=(x_3-x_4)^2+(y_4-y_3)^2$$

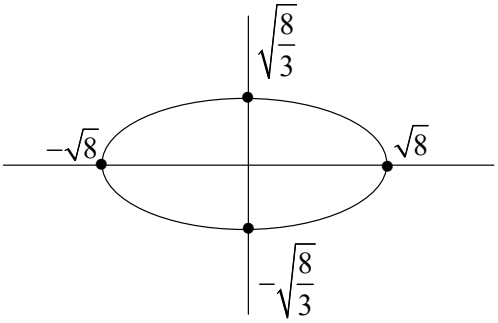
$$=CD^2$$

Thus, ABCD is a parallelogram AB = CD . Hence , it is a rectangle.

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66. $\{-1,0,1\}$

$R=\{(x,y):x,y\in z,x^2+3y^2\le 8\}$



For domain of R^{-1}

Collection of all integral of y's

For $x=0, 3y^2\le 8$

$\Rightarrow y\in\{-1,0,1\}$

67. We have, $\sum_{i=1}^{9}(x_1-5)=9$ *and* $\sum_{i=1}^{9}(x_1-5)^2=45$

$$SD=\sqrt{\frac{\sum_{i=1}^{9}(x_1-5)^2}{9}-\left(\frac{\sum_{i=1}^{9}(x_1-5)}{9}\right)^2}$$

$$\Rightarrow SD=\sqrt{\frac{45}{9}-\left(\frac{9}{9}\right)^2}$$

$$\Rightarrow SD=\sqrt{5-1}=\sqrt{4}=2$$

68. Given integral $=\int_0^1 0dx+\int_1^2 1\{x\}dx+\int_2^3 2^3\{x\}dx+.....+\int_9^{10}9^3\{x\}dx$

$$=\left(1^3+2^3+.....9^3\right)\int_0^1\{x\}dx=\left(\frac{9\times 10}{2}\right)^2\times\int_0^1 xdx=\frac{2025}{2}$$

69. $z=1+\cos\frac{8\pi}{5}+i\sin\frac{8\pi}{5}$

$$=2\cos^2\frac{4\pi}{5}+i2\sin\frac{4\pi}{5}\cos\frac{4\pi}{5}$$

$$=2\cos\frac{4\pi}{5}\left(\cos\frac{4\pi}{5}+i\sin\frac{4\pi}{5}\right)$$

$$=-2\cos\frac{\pi}{5}\left(-\cos\frac{\pi}{5}+i\sin\frac{\pi}{5}\right)$$

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$$= 2\cos\frac{\pi}{5}\left(\cos\frac{\pi}{5} - i\sin\frac{\pi}{5}\right)$$

$$= 2\cos\frac{\pi}{5}\left(\cos\left(-\frac{\pi}{5}\right) + i\sin\left(-\frac{\pi}{5}\right)\right)$$

Thus, $|z| = 2\cos\frac{\pi}{5}$ *and* $\arg(z) = -\frac{\pi}{5}$

70. Let A_k be the of positive integers from 1 to 1000, which are divisible by k.
If $[.]$ denotes the greatest integer function, then

$$n(A_2) = \left[\frac{1000}{2}\right] = 500$$

$$n(A_3) = \left[\frac{1000}{3}\right] = 333$$

$$n(A_5) = \left[\frac{1000}{5}\right] = 200$$

Also $n(A_2 \cap A_3) = \left[\frac{1000}{6}\right] = 166$

$$n(A_3 \cap A_5) = \left[\frac{1000}{15}\right] = 66$$

$$n(A_2 \cap A_5) = \left[\frac{1000}{10}\right] = 100$$

and $n(A_2 \cap A_3 \cap A_5) = \left[\frac{1000}{30}\right] = 33$

So, $n(A_2 \cap A_3 \cap A_5) = 500 + 333 + 200 - 166 - 66 - 100 + 33 = 734$

71. Let $\hat{i}, \hat{j}, \hat{k}$ be the unit vectors along OA,OB,OC. CN,BM are diagonals skew to OA
Let 1 be the S,D between OA and CN let $\overrightarrow{OA} = x\hat{i}, \overrightarrow{OB} = y\hat{j}, \overrightarrow{OC} = z\hat{k} \Rightarrow \overrightarrow{ON} = x\hat{i} + y\hat{j}$
Equation of OA and CN are $\vec{r} = x\hat{i}$ and $\hat{r} = z\hat{k} + p\left(x\hat{i} + y\hat{j} - z\hat{k}\right)$ respectively.
The shortest distance between those two lines.

$$\ell = \frac{yz}{\sqrt{y^2 + z^2}} \Rightarrow \frac{1}{\ell^2} + \frac{1}{y^2} + \frac{1}{z^2}$$

Similarly $\frac{1}{m^2} = \frac{1}{z^2} + \frac{1}{x^2}, \frac{1}{n^2} = \frac{1}{x^2} + \frac{1}{y^2}, \Rightarrow \frac{1}{\ell^2} + \frac{1}{m^2} + \frac{1}{n^2} = 2\left(\frac{1}{OA^2} + \frac{1}{OB^2} + \frac{1}{OC^2}\right)$

72. $|A| = (2k+1)^3, |B| = 0$ (since B is a skew – symmetric matrix of order 3)

$$\Rightarrow \det(adj\, A) = |A|^{n-1} = \left((2k+1)^3\right)^2 = 10^6$$

$$\Rightarrow 2k + 1 = 10 \text{ or } 2k = 9$$

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$\Rightarrow [k]=4.$

73. Area of the shaded region is

$=\frac{1}{2}\times 8\times 8-\frac{2}{3}\times 32$

$=\frac{1}{3}\times 32$

$\therefore \frac{32}{3}=\Delta$

$\therefore [\Delta]=10$

74. The equation of the line through P(x,y) making an angle with the x-axis which is suplementary to the angle made by the tangent at P(x,y) is

$Y-y=-\frac{dy}{dx}(X-x).....(1)$

At the point where it meets the x-axis

$Y=0, X=x+\frac{y}{\frac{dy}{dx}}\Rightarrow OA=x+\frac{y}{\frac{dy}{dx}}.....(2)$

The line through P(x,y) and perpendicular to (1) is

$Y-y=\frac{dx}{dy}(X-x).$

At the point where it meets the y-axis

$X=0, Y=y-\frac{x}{\frac{dy}{dx}}\Rightarrow OB=y-\frac{x}{\frac{dy}{dx}}$

$\text{Since}\ \ OA=OB,$

$x+\frac{y}{\frac{dy}{dx}}=y-\frac{x}{\frac{dy}{dx}}\Rightarrow (y-x)=\frac{y+x}{\frac{dy}{dx}}\Rightarrow \frac{dy}{dx}=\frac{y+x}{y-x}$

Writing y = vx this equation becomes

$v+x\frac{dv}{dx}=\frac{1+v}{v-1}\ \ \Rightarrow x\frac{dv}{dx}=\frac{1+2v-v^2}{v-1}$

$\Rightarrow \frac{(1-v)dv}{1+2v-v^2}+\frac{dx}{x}=0$

$\Rightarrow \ell n\left(12v-v^2\right)+\ell n\ X^2=\text{constant}$

$\Rightarrow x^2+2xy-y^2=c$

Since the curve passes through $(1,1), c=2$

Hence , the required curve is $x^2-y^2+2xy=2.$

75. $\frac{x}{y}=m$

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Let $x = my$ be the tangent Applying

$$\frac{|3-4m|}{\sqrt{1+m^2}} = 2$$

$$12m^2 - 24m + 5 = 0$$

$$(m_1 + m_2)^2 - 2m_1m_2 = \left(\frac{24}{12}\right)^2 - 2 \times \frac{5}{12}$$

$$= \frac{19}{6}$$

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